Appendix

Semiconductors and CMOS Integrated Circuits

Semiconductors are formed by doping a thin slice of a pure silicon crystal with a small amount of a dopant that fits relatively easily into the crystalline structure of the silicon. Dopants are differentiated on the basis of whether they have either three valence electrons or five valence electrons. A silicon crystalline structure is such that each silicon atom shares its four valence electrons with its four nearest neighbors, thereby completing its valence structure. The atoms of a dopant with five valence electrons, referred to as a *n*-type dopant, fit in the physical structure of the crystal, but their fifth electrons are held only loosely by their parent atoms in the bonded structure. Consequently, an applied electric field can cause such electrons to flow as a current. On the other hand, a dopant atom with only three valence electrons, a *p*-type dopant, has a vacant valence site. Under the influence of an applied electric field, an electron from a neighboring silicon atom in the bonded structure can jump from its host and fill a vacant dopant site, leaving behind a vacancy at its host. This migration, visualized as a leapfrogging of electrons from hole to hole, establishes a current.

Current is due to the movement of electrons, which are negative charge carriers. Current is measured, however, in the opposite direction of flow, by convention—since the days of Benjamin Franklin. (Think of current as being the motion of an equivalent positive charge moving in the opposite direction of an electron, whose charge is negative). Holes move in the direction of current, although the underlying physical movement of electrons is in the opposite direction. Thermal agitation causes both types of charge carriers to be present in a semiconductor. If the majority carrier is a hole, the device is said to be a *p*-type device; if the majority carrier is an electron, the device is said to be an *n*-type device. Bipolar transistors rely on both types of carriers. Metal-oxide silicon semiconductors rely on a majority carrier, either an electron or a hole, but not both. The type and relative amount of dopant determine the type of a semiconductor material.

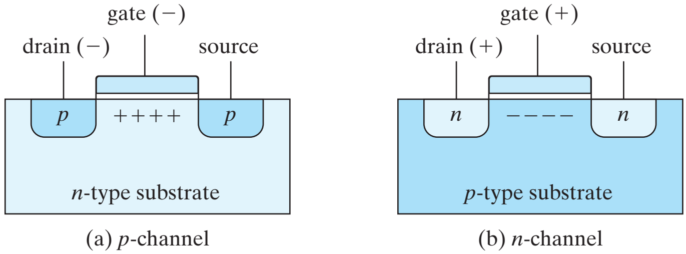


FIGURE A.1
Basic structure of MOS transistor

The basic structure of a metal-oxide semiconductor (MOS) transistor is shown in Fig. A.1. The *p*-channel MOS transistor consists of a lightly doped substrate of *n*-type silicon material. Two regions are heavily doped with *p*-type impurities by a diffusion process to form the *source* and *drain*. The source terminal supplies charge carriers to an external circuit; the drain terminal removes charge carriers from the circuit. The region between the two *p*-type sections serves as the *channel*. In its simplest form, the gate is a metal plate separated from the channel by an insulated dielectric of silicon dioxide. A negative voltage (with respect to the substrate) at the gate terminal causes an induced electric field in the channel that attracts *p*-type carriers (holes) from the substrate. As the magnitude of the negative voltage increases, the region below the gate accumulates more positive carriers, the conductivity increases, and current can flow from source to drain, provided that a voltage difference is maintained between these two terminals.

There are four basic types of MOS structures. The channel can be *p* or *n* type, depending on whether the majority carriers are holes or electrons. The mode of operation can be enhancement or depletion, depending on the state of the channel region at zero gate voltage. If the channel is initially doped lightly with *p*-type impurity (in which case it is called a *diffused channel*), a conducting channel exists at zero gate voltage and the device is said to operate in the *depletion* mode. In this mode, current flows unless the channel is depleted by an applied gate field. If the region beneath the gate is left initially uncharged, a channel must be induced by the gate field before current can flow. Thus, the channel current is enhanced by the gate voltage, and such a device is said to operate in the *enhancement* mode.

The source is the terminal through which the majority carriers enter the device. If the majority carrier is a hole (*p*-type channel), the source terminal supplies current to the circuit; if the majority carrier is an electron (*n*-type channel), the source removes current from the circuit. The drain is the terminal through which the majority carriers leave the device. In a *p*-channel MOS, the source terminal is connected to the substrate and a negative voltage is applied to the drain terminal. When the gate voltage is above a threshold voltage V_T (about -2 V), no current flows in the channel and the drain-to-source path is like an open circuit. When the gate voltage is sufficiently negative below V_T, a channel is formed and *p*-type carriers flow from source to drain. *p*-type carriers are positive and correspond to a positive current flow from source to drain.

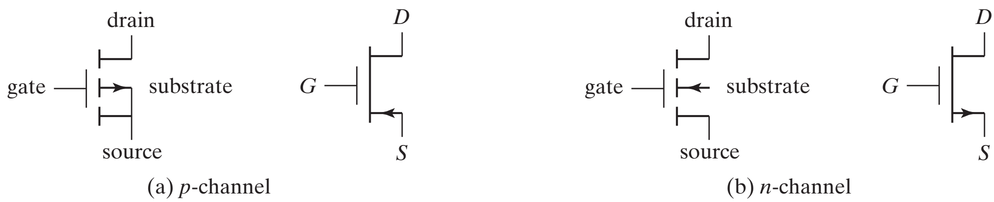


FIGURE A.2
Symbols for MOS transistors

In the *n*-channel MOS, the source terminal is connected to the substrate and a positive voltage is applied to the drain terminal. When the gate voltage is below the threshold voltage V_T (about 2 V), no current flows in the channel. When the gate voltage is sufficiently positive above V_T to form the channel, *n*-type carriers flow from source to drain. *n*-type carriers are negative and correspond to a positive current flow from drain to source. The threshold voltage may vary from 1 to 4 V, depending on the particular process used.

The graphic symbols for the MOS transistors are shown in Fig. A.2. The symbol for the enhancement type is the one with the broken-line connection between source and drain. In this symbol, the substrate can be identified and is shown connected to the source. An alternative symbol omits the substrate, and instead an arrow is placed in the source terminal to show the direction of *positive* current flow (from source to drain in the *p*-channel MOS and from drain to source in the *n*-channel MOS).

Because of the symmetrical construction of source and drain, the MOS transistor can be operated as a bilateral device. Although normally operated so that carriers flow from source to drain, there are circumstances when it is convenient to allow carriers to flow from drain to source.

One advantage of the MOS device is that it can be used not only as a transistor, but as a resistor as well. A resistor is obtained from the MOS by permanently biasing the gate terminal for conduction. The ratio of the source–drain voltage to the channel current then determines the value of the resistance. Different resistor values may be constructed during manufacturing by fixing the channel length and width of the MOS device.

Three logic circuits using MOS devices are shown in Fig. A.3. For an *n*-channel MOS, the supply voltage V_{DD} is positive (about 5 V), to allow positive current flow from drain to source. The two voltage levels are a function of the threshold voltage V_T. The low level is anywhere from zero to V_T, and the high level ranges from V_T to V_{DD}. The *n*-channel gates usually employ positive logic. The *p*-channel MOS circuits use a negative voltage for V_{DD}, to allow positive current flow from source to drain. The two voltage levels are both negative above and below the negative threshold voltage V_T. *p*-channel gates usually employ negative logic.

The inverter circuit shown in Fig. A.3(a) uses two MOS devices. *Q1* acts as the load resistor and *Q2* as the active device. The load-resistor MOS has its gate con-

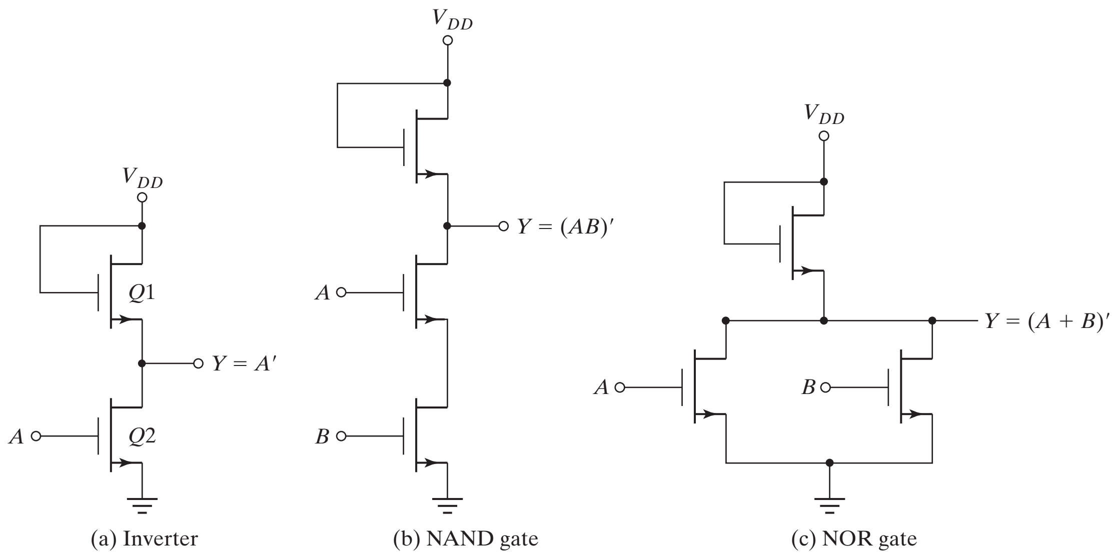


FIGURE A.3
n-channel MOS logic circuits

nected to V_{DD}, thus maintaining it in the conduction state. When the input voltage is low (below V_T), *Q2* turns off. Since *Q1* is always on, the output voltage is about V_{DD}. When the input voltage is high (above V_T), *Q2* turns on. Current flows from V_{DD} through the load resistor *Q1* and into *Q2*. The geometry of the two MOS devices must be such that the resistance of *Q2*, when conducting, is much less than the resistance of *Q1* to maintain the output *Y* at a voltage below V_T.

The NAND gate shown in Fig. A.3(b) uses transistors in series. Inputs *A* and *B* must both be high for all transistors to conduct and cause the output to go low. If either input is low, the corresponding transistor is turned off and the output is high. Again, the series resistance formed by the two active MOS devices must be much less than the resistance of the load-resistor MOS. The NOR gate shown in Fig. A.3(c) uses transistors in parallel. If either input is high, the corresponding transistor conducts and the output is low. If all inputs are low, all active transistors are off and the output is high.

A.1 COMPLEMENTARY MOS

Complementary MOS (CMOS) circuits take advantage of the fact that both *n*-channel and *p*-channel devices can be fabricated on the same substrate. CMOS circuits consist of both types of MOS devices, interconnected to form logic functions. The basic circuit is the inverter, which consists of one *p*-channel transistor and one *n*-channel transistor, as shown in Fig. A.4(a). The source terminal of the *p*-channel device is at V_{DD}, and the source terminal of the *n*-channel device is at ground. The value of V_{DD} may be anywhere

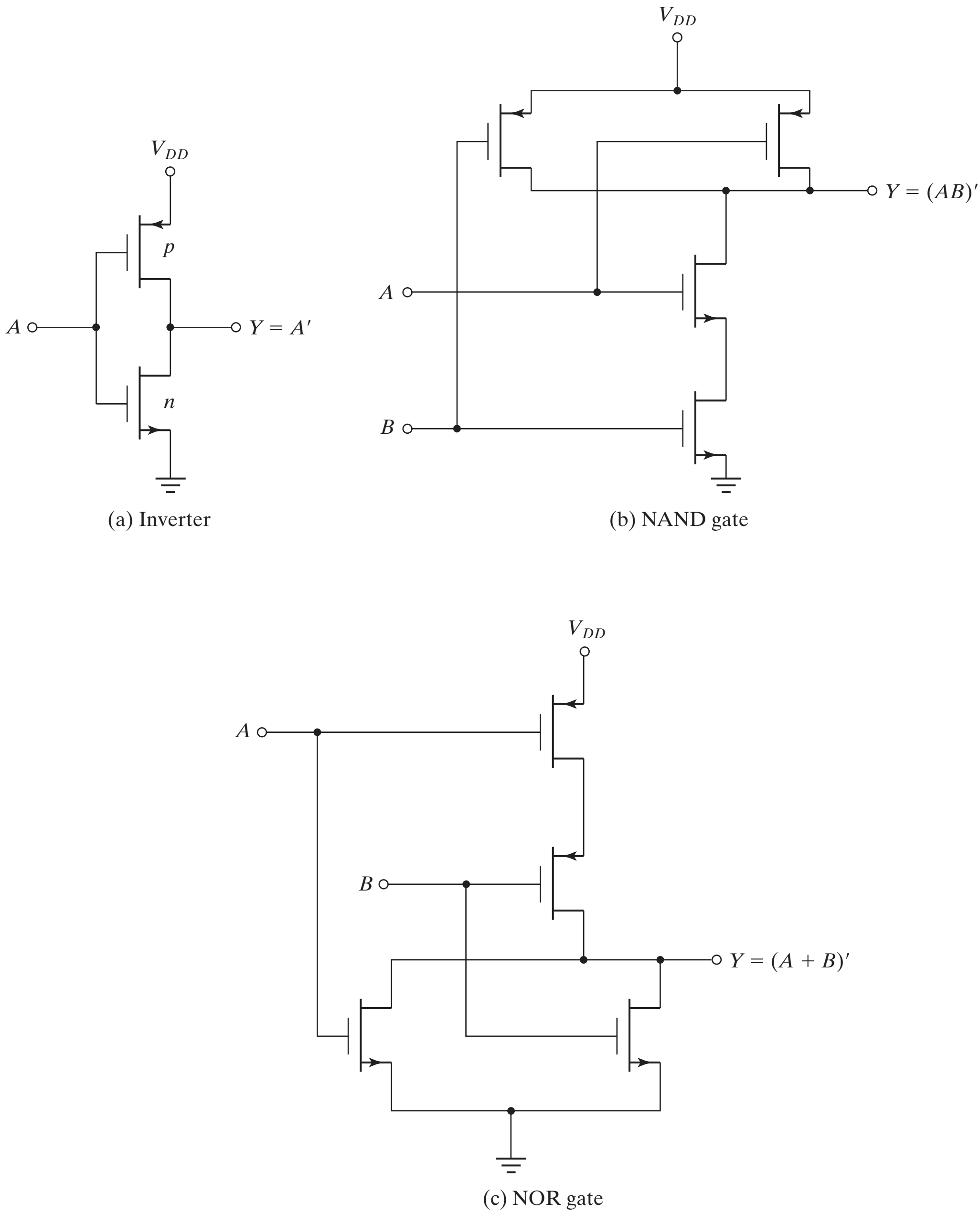


FIGURE A.4
CMOS logic circuits

from +3 to +18 V. The two voltage levels are 0 V for the low level and V_{DD} for the high level (typically, 5 V).

To understand the operation of the inverter, we must review the behavior of the MOS transistor from the previous section:

1. The *n*-channel MOS conducts when its gate-to-source voltage is positive.

2. The *p*-channel MOS conducts when its gate-to-source voltage is negative.

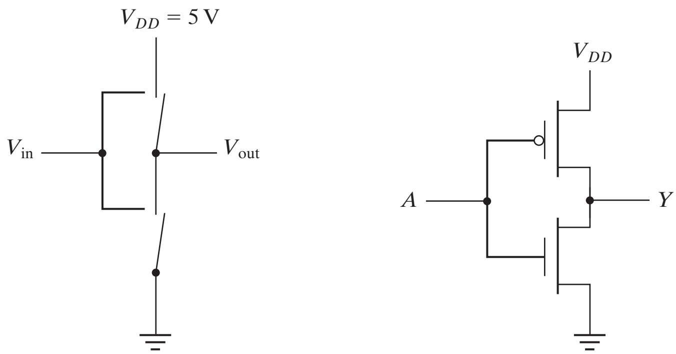
3. Either type of device is turned off if its gate-to-source voltage is zero.

Now consider the operation of the inverter. When the input is low, both gates are at zero potential. The input is at $-V_{DD}$ relative to the source of the *p*-channel device and at 0 V relative to the source of the *n*-channel device. The result is that the *p*-channel device is turned on and the *n*-channel device is turned off. Under these conditions, there is a low-impedance path from V_{DD} to the output and a very high impedance path from output to ground. Therefore, the output voltage approaches the high level V_{DD} under normal loading conditions. When the input is high, both gates are at V_{DD} and the situation is reversed: The *p*-channel device is off and the *n*-channel device is on. The result is that the output approaches the low level of 0 V.

Two other CMOS basic gates are shown in Fig. A.4. A two-input NAND gate consists of two *p*-type units in parallel and two *n*-type units in series, as shown in Fig. A.4(b). If all inputs are high, both *p*-channel transistors turn off and both *n*-channel transistors turn on. The output has a low impedance to ground and produces a low state. If any input is low, the associated *n*-channel transistor is turned off and the associated *p*-channel transistor is turned on. The output is coupled to V_{DD} and goes to the high state. Multiple-input NAND gates may be formed by placing equal numbers of *p*-type and *n*-type transistors in parallel and series, respectively, in an arrangement similar to that shown in Fig. A.4(b).

A two-input NOR gate consists of two *n*-type units in parallel and two *p*-type units in series, as shown in Fig. A.4(c). When all inputs are low, both *p*-channel units are on and both *n*-channel units are off. The output is coupled to V_{DD} and goes to the high state. If any input is high, the associated *p*-channel transistor is turned off and the associated *n*-channel transistor turns on, connecting the output to ground and causing a low-level output.

MOS transistors can be considered to be electronic switches that either conduct or are open. As an example, the CMOS inverter can be visualized as consisting of two switches as shown in Fig. A.5(a). Applying a low voltage to the input causes the upper switch (*p*) to close, supplying a high voltage to the output. Applying a high voltage to the input causes the lower switch (*n*) to close, connecting the output to ground. Thus, the output V_{out} is the complement of the input V_{in}. Commercial applications often use other graphic symbols for MOS transistors to emphasize the logical behavior of the switches. The arrows showing the direction of current flow are omitted. Instead, the gate input of the *p*-channel transistor is drawn with an inversion bubble on the gate terminal to show that it is enabled with a low voltage. The inverter circuit is redrawn with these symbols in Fig. A.5(b). A logic 0 in the input causes the upper transistor to conduct, making the output logic 1. A logic 1 in the input enables the lower transistor, making the output logic 0.



(a) Switch model (b) Logical model

FIGURE A.5
CMOS inverter

CMOS Characteristics

When a CMOS logic circuit is in a static state, its power dissipation is very low. This is because at least one transistor is always off in the path between the power supply and ground when the state of the circuit is not changing. As a result, a typical CMOS gate has static power dissipation on the order of 0.01 mW. However, when the circuit is changing state at the rate of 1 MHz, the power dissipation increases to about 1 mW, and at 10 MHz it is about 5 mW.

CMOS logic is usually specified for a single power-supply operation over a voltage range from 3 to 18 V with a typical V_{DD} value of 5 V. Operating CMOS at a larger power-supply voltage reduces the propagation delay time and improves the noise margin, but the power dissipation is increased. The propagation delay time with $V_{DD} = 5$ V ranges from 5 to 20 ns, depending on the type of CMOS used. The noise margin is usually about 40% of the power supply voltage. The fan-out of CMOS gates is about 30 when they are operated at a frequency of 1 MHz. The fan-out decreases with an increase in the frequency of operation of the gates.

There are several series of the CMOS digital logic family. The 74C series are pin and function compatible with TTL devices having the same number. For example, CMOS IC type 74C04 has six inverters with the same pin configuration as TTL type 7404. The high-speed CMOS 74HC series is an improvement over the 74C series, with a tenfold increase in switching speed. The 74HCT series is electrically compatible with TTL ICs. This means that circuits in this series can be connected to inputs and outputs of TTL ICs without the need of additional interfacing circuits. Newer versions of CMOS are the high-speed series 74VHC and its TTL-compatible version 74VHCT.

The CMOS fabrication process is simpler than that of TTL and provides a greater packing density. Thus, more circuits can be placed on a given area of silicon at a reduced cost per function. This property, together with the low power dissipation of CMOS circuits, good noise immunity, and reasonable propagation delay, makes CMOS the most popular standard as a digital logic family.

A.2 CMOS TRANSMISSION GATE CIRCUITS

A special CMOS circuit that is not available in the other digital logic families is the *transmission gate*. The transmission gate is essentially an electronic switch that is controlled by an input logic level. It is used to simplify the construction of various digital components when fabricated with CMOS technology.

Figure A.6(a) shows the basic circuit of the transmission gate. Whereas a CMOS inverter consists of a p-channel transistor connected in series with an n-channel transistor, a transmission gate is formed by one n-channel and one p-channel MOS transistor connected in parallel.

The n-channel substrate is connected to ground and the p-channel substrate is connected to V_{DD}. When the N gate is at V_{DD} and the P gate is at ground, both transistors conduct and there is a closed path between input X and output Y. When the N gate is at ground and the P gate is at V_{DD}, both transistors are off and there is an open circuit between X and Y. Figure A.6(b) shows the block diagram of the transmission gate. Note that the terminal of the p-channel gate is marked with the negation symbol. Figure A.6(c) demonstrates the behavior of the switch in terms of positive-logic assignment with V_{DD} equivalent to logic 1 and ground equivalent to logic 0.

The transmission gate is usually connected to an inverter, as shown in Fig. A.7. This type of arrangement is referred to as a *bilateral switch*. The control input C is connected directly to the n-channel gate and its inverse to the p-channel gate. When $C = 1$, the

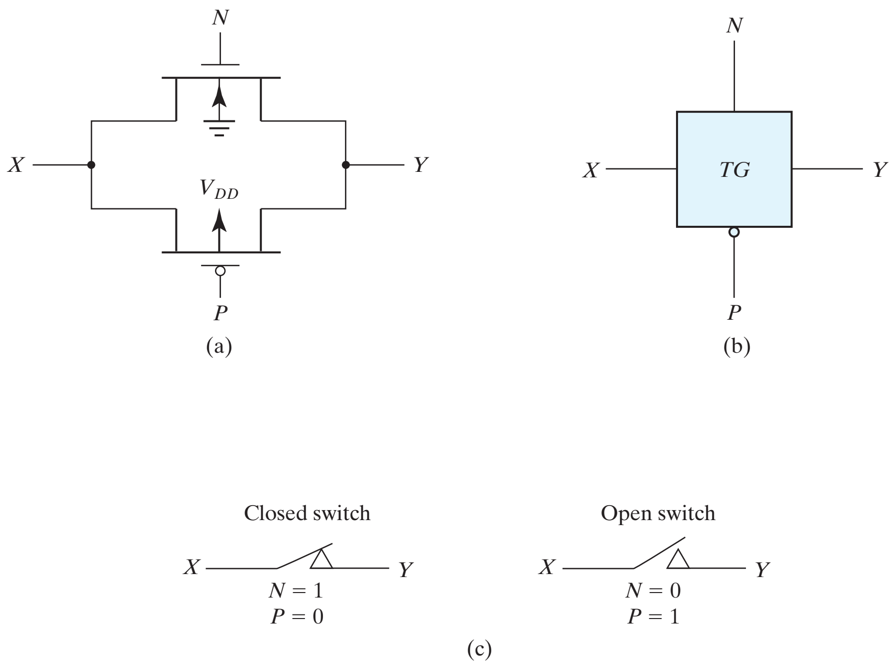


FIGURE A.6
Transmission gate (TG)

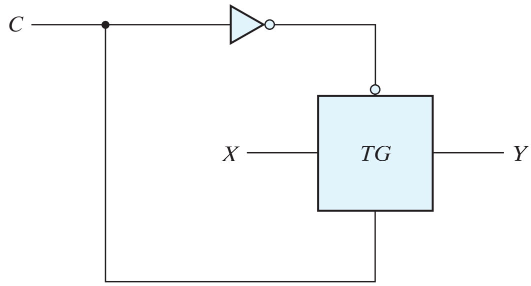


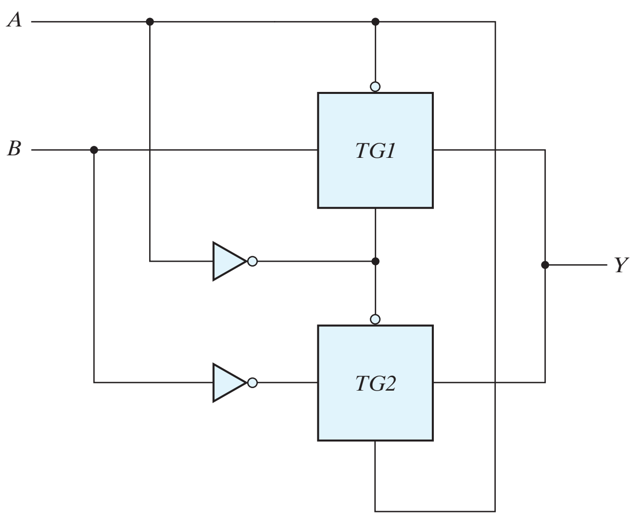
FIGURE A.7
Bilateral switch

switch is closed, producing a path between X and Y. When $C = 0$, the switch is open, disconnecting the path between X and Y.

Various circuits can be constructed that use the transmission gate. To demonstrate its usefulness as a component in the CMOS family, we will show three examples.

The exclusive-OR gate can be constructed with two transmission gates and two inverters, as shown in Fig. A.8. Input A controls the paths in the transmission gates and input B is connected to output Y through the gates. When input A is equal to 0, transmission gate *TG1* is closed and output Y is equal to input B. When input A is equal to 1, *TG2* is closed and output Y is equal to the complement of input B. This results in the exclusive-OR truth table, as indicated in Fig. A.8.

Another circuit that can be constructed with transmission gates is the multiplexer. A four-to-one-line multiplexer implemented with transmission gates is shown in Fig. A.9. The *TG* circuit provides a transmission path between its horizontal input and output lines



A	*B*	*TG1*	*TG2*	*Y*
0	0	close	open	0
0	1	close	open	1
1	0	open	close	1
1	1	open	close	0

FIGURE A.8
Exclusive-OR constructed with transmission gates

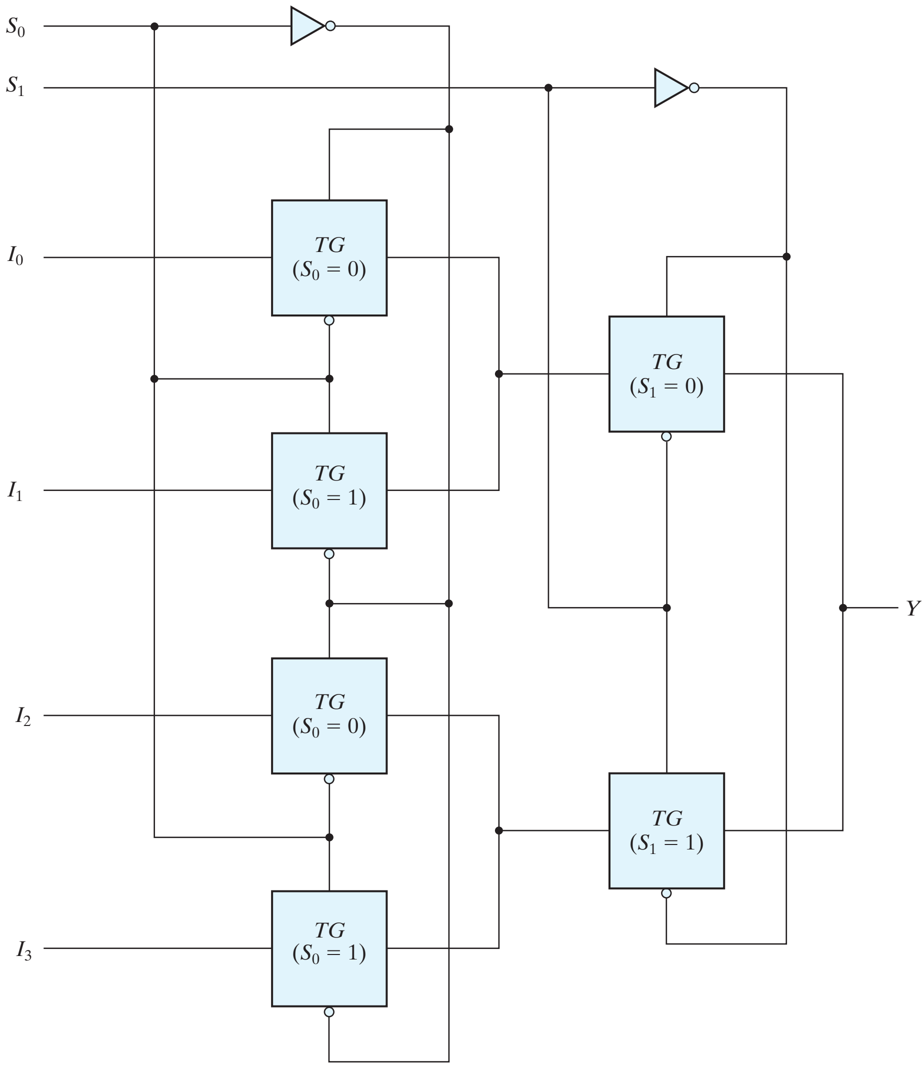


FIGURE A.9
Multiplexer with transmission gates

when the two vertical control inputs have the value of 1 in the uncircled terminal and 0 in the circled terminal. With an opposite polarity in the control inputs, the path disconnects and the circuit behaves like an open switch. The two selection inputs, S_1 and S_0, control the transmission path in the *TG* circuits. Inside each box is marked the condition for the transmission gate switch to be closed. Thus, if $S_0 = 0$ and $S_1 = 0$, there is a closed path from input I_0 to output Y through the two *TG*s marked with $S_0 = 0$ and $S_1 = 0$. The other three inputs are disconnected from the output by one of the other *TG* circuits.

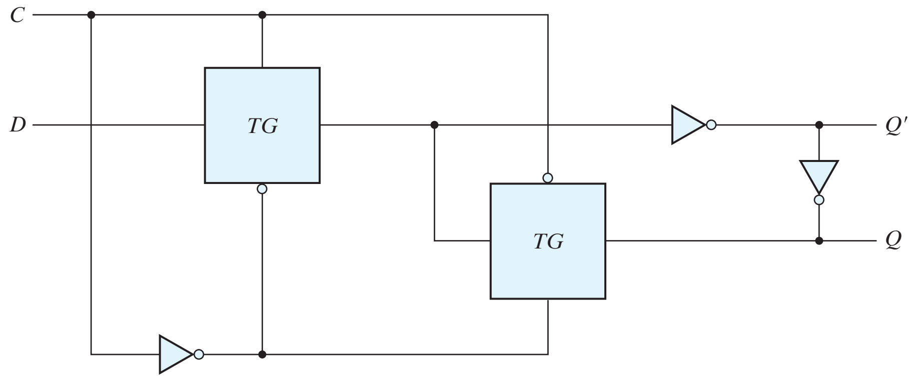


FIGURE A.10
Gated *D* latch with transmission gates

The level-sensitive D flip-flop commonly referred to as the gated D latch can be constructed with transmission gates, as shown in Fig. A.10. The C input controls two transmission gates TG. When $C = 1$, the TG connected to input D has a closed path and the one connected to output Q has an open path. This configuration produces an equivalent circuit from input D through two inverters to output Q. Thus, the output follows the data input as long as C remains active. When C switches to 0, the first TG disconnects input D from the circuit and the second TG produces a closed path between the two inverters at the output. Thus, the value that was present at input D at the time that C went from 1 to 0 is retained at the Q output.

A master–slave D flip-flop can be constructed with two circuits of the type shown in Fig. A.10. The first circuit is the master and the second is the slave. Thus, a master–slave D flip-flop can be constructed with four transmission gates and six inverters.

A.3 SWITCH-LEVEL MODELING WITH HDL

CMOS is the dominant digital logic family used with integrated circuits. By definition, CMOS is a complementary connection of an NMOS and a PMOS transistor. MOS transistors can be considered to be electronic switches that either conduct or are open. By specifying the connections among MOS switches, the designer can describe a digital circuit constructed with CMOS. This type of description is called *switch-level modeling* in Verilog HDL.

The two types of MOS switches are specified in Verilog HDL with the keywords **nmos** and **pmos**. They are instantiated by specifying the three terminals of the transistor, as shown in Fig. A.2:

```
nmos (drain, source, gate);
pmos (drain, source, gate);
```

Switches are considered to be primitives, so the use of an instance name is optional.

The connections to a power source (V_{DD}) and to ground must be specified when MOS circuits are designed. Power and ground are defined with the keywords **supply1** and **supply0**. They are specified, for example, with the following statements:

```
supply1 PWR;
supply0 GRD;
```

Sources of type **supply1** are equivalent to V_{DD} and have a value of logic 1. Sources of type **supply0** are equivalent to ground connection and have a value of logic 0.

The description of the CMOS inverter of Fig. A.4(a) is shown in HDL Example A.1. The input, the output, and the two supply sources are declared first. The module instantiates a PMOS and an NMOS transistor. The output *Y* is common to both transistors at their drain terminals. The input is also common to both transistors at their gate terminals. The source terminal of the PMOS transistor is connected to PWR and the source terminal of the NMOS transistor is connected to GRD.

HDL Example A.1

```
// CMOS inverter of Fig. A.4(a)
module inverter (Y, A);
  input A;
  output Y;
  supply1 PWR;
  supply0 GRD;
  pmos (Y, PWR, A);                // (Drain, source, gate)
  nmos (Y, GRD, A);                // (Drain, source, gate)
endmodule
```

The second module, set forth in HDL Example A.2, describes the two-input CMOS NAND circuit of Fig. A.4(b). There are two PMOS transistors connected in parallel, with their source terminals connected to PWR. There are also two NMOS transistors connected in series and with a common terminal *W1*. The drain of the first NMOS is connected to the output, and the source of the second NMOS is connected to GRD.

HDL Example A.2

```
// CMOS two-input NAND of Fig. A.4(b)
module NAND2 (Y, A, B);
  input A, B;
  output Y;
  supply1 PWR;
  supply0 GRD;
  wire W1;                         // terminal between two nmos
  pmos (Y, PWR, A);                // source connected to Vdd
  pmos (Y, PWR, B);                // parallel connection
```

```
  nmos (Y, W1, A);                    // serial connection
  nmos (W1, GRD, B);      // source connected to ground
endmodule
```

Transmission Gate

The transmission gate is instantiated in Verilog HDL with the keyword **cmos**. It has an output, an input, and two control signals, as shown in Fig. A.6. It is referred to as a **cmos** switch. The relevant code is as follows:

```
cmos (output, input, ncontrol, pcontrol); // general description
cmos (Y, X, N, P); // transmission gate of Fig. A.6(b)
```

Normally, ncontrol and pcontrol are the complement of each other. The **cmos** switch does not need power sources, since V_{DD} and ground are connected to the substrates of the MOS transistors. Transmission gates are useful for building multiplexers and flip-flops with CMOS circuits.

HDL Example A.3 describes a circuit with **cmos** switches. The exclusive-OR circuit of Fig. A.8 has two transmission gates and two inverters. The two inverters are instantiated within the module describing a CMOS inverter. The two **cmos** switches are instantiated without an instance name, since they are primitives in the language. A test module is included to test the circuit's operation. Applying all possible combinations of the two inputs, the result of the simulator verifies the operation of the exclusive-OR circuit. The output of the simulation is as follows:

$$A = 0 \quad B = 0 \quad Y = 0$$
$$A = 0 \quad B = 1 \quad Y = 1$$
$$A = 1 \quad B = 0 \quad Y = 1$$
$$A = 1 \quad B = 1 \quad Y = 0$$

HDL Example A.3

```
//CMOS_XOR with CMOS switches, Fig. A.8
module CMOS_XOR (A, B, Y);
  input A, B;
  output Y;
  wire A_b, B_b;
 // instantiate inverter
 inverter v1 (A_b, A);
 inverter v2 (B_b, B);
 // instantiate cmos switch
 cmos (Y, B, A_b, A);                //(output, input, ncontrol, pcontrol)
 cmos (Y, B_b, A, A_b);
endmodule
```

```
// CMOS inverter Fig. A.4(a)
module inverter (Y, A);
  input A;
  output Y;
  supply1 PWR;
  supply0 GND;
  pmos (Y, PWR, A);                  //(Drain, source, gate)
  nmos (Y, GND, A);                  //(Drain, source, gate)
endmodule
// Stimulus to test CMOS_XOR
module  test_CMOS_XOR;
  reg  A,B;
  wire  Y;
 //Instantiate CMOS_XOR
 CMOS_XOR X1 (A, B, Y);
 // Apply truth table
  initial
    begin
     A = 1'b0; B = 1'b0;
 #5 A = 1'b0; B = 1'b1;
 #5 A = 1'b1; B = 1'b0;
 #5 A = 1'b1; B = 1'b1;
end
// Display results
initial
 $monitor  ("A=%b B= %b Y =%b", A, B, Y);
endmodule
```

WEB SEARCH TOPICS

Conductor
Semiconductor
Insulator
Electrical properties of materials
Valence electron
Diode
Transistor
CMOS process
CMOS logic gate
CMOS inverter